# MYSX-256/512: A 256-bit Block Cipher with 512-bit Keys (v0.1-draft)

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**Abstract**

We present MYSX-256/512, a 256-bit block cipher with 512-bit keys and 128 rounds. The design combines Feistel cells with Simon's round function, a Myszkowski byte transposition, word rotations, and an ARX key schedule. We provide the full specification, test vectors, and a self-analysis covering differential, linear, integral, and related-key cryptanalysis. The differential analysis reports an observed upper bound on the best single-trail weight (114 at 14 rounds, extrapolating to the 128-round threshold at r≈15). The r=4 trail has been directly verified at w=32 with a clustering factor of 2.00× (observed $2^{-7}$ vs MILP $2^{-8}$). The linear analysis is built on an exact model of F: the LAT admits a closed form (`|corr(b,g)| = 2^{-h(b)}`, `h(b) = rank(A_b)/2` on a 32-cycle), mask propagation is bit-exact, and every trail carries an exact key-dependent sign. Within this model we verify `|corr|=1` chains over up to 9 consecutive rounds (three independent keys), and exact linear hulls at r=10,11,12 match the real cipher within sampling error ($|z| < 0.4$). Beam search over single-bit starts gives upper bounds on the best single-trail weight up to 14 rounds (weight 12 at r=14, window r=12). No lower bound over 128 rounds is established; the weight-0 windows are a property of 5–9 round windows, not an attack on the full cipher. A preliminary integral analysis with standard division-property rules finds integral distinguishers covering 6 rounds with a $2^{32}$-text integral (word 0 active) and 5 rounds with a $2^{64}$-text integral (words 0 and 4 active); a 10-round single-bit dependency propagation is also reported but is not an integral distinguisher (2 texts). Only r0=0 was tested. The key schedule was audited with 22 structural and empirical tests up to weight-6 with no persistent Δrk=0 pattern found. MYSX has not been externally audited. It is released as v0.1-draft for public review and must not be used to protect real data.

## 1. Introduction

MYSX-256/512 is an experimental block cipher. It is offered as a public proposal for cryptanalysis, not as a production primitive. The author makes no claim of security beyond the observed results reported here, and explicitly welcomes external review.

### 1.1. Security objectives (and non-objectives)

**Model.** MYSX is analyzed in the single-key, chosen-plaintext model for confidentiality, combined with an encrypt-then-MAC AEAD wrapper for integrity (Section 2.9). The

design does not claim security under related-key, chosen-ciphertext, or side-channel models; see Section 5 for related-key coverage and Section 9 for engineering notes.

**Claimed security level.** None formally. The paper reports observed upper bounds on single-trail weights (Section 3), a verified exact linear model up to a small number of rounds (Section 4), and preliminary integral distinguishers at 5–6 rounds for multi-bit inputs (§4.9). These are inputs to cryptanalysis, not proofs.

**Non-objectives.** MYSX does not claim:

- post-quantum security beyond the trivial Grover bound ($2^{256}$ quantum iterations for a 512-bit key);
- resistance to related-key attacks beyond the heuristic coverage in Section 5;
- resistance to integral attacks beyond the preliminary 5–6 round distinguishers reported in §4.9, nor resistance to impossible- differential, boomerang, or algebraic attacks (none analyzed);
- constant-time execution in JavaScript (side channels out of scope).

## 1.2. Design rationale

MYSX combines four building blocks:

- **Simon's F** `F(x) = (rotl(x,1) ∧ rotl(x,8)) ⊕ rotl(x,2)`. Chosen for its **studied** diffusion/linear/differential properties in Simon and its simplicity (no S-boxes, only AND/XOR/rotl). It does **not** inherit Simon's security proofs; the surrounding structure is different. See Section 7 for what transfers and what does not.
- **Feistel cells over 32-bit word pairs.** Four parallel cells per round, then a byte transposition over the full 32-byte state, then word rotations. The result is classified as an SPN-like structure with Feistel cells, not as a "2-branch Feistel". See Section 2.1.
- **Myszkowski transposition** (Section 2.3.1). A fixed byte permutation derived from the keyword `MYSZKOWS`. The choice is didactic; it provides byte-level diffusion but no cryptographic argument beyond that. An optimized permutation would provide strictly better diffusion per round; see Section 9.
- **ARX key schedule** (Section 2.4). A bijective Feistel 8+8 update with modular addition, XOR, and rotations. Chosen for simplicity and to avoid S-boxes; analyzed in Section 5.

The parameter choices (256-bit block, 512-bit key, 128 rounds) are not derived from a formal argument. They were chosen to:

- provide a large block (avoid birthday-bound issues of 64-bit blocks);
- provide a large key (Grover leaves 256-bit security);
- provide a large round count relative to Simon128/128 (68 rounds) as a precaution against unknown trails.

No claim is made that 128 rounds is sufficient. The round count is a parameter, not a security level.

### 1.3. Comparison with related designs

| Design | Block | Key | Rounds | Structure | S-boxes |
|---|---|---|---|---|---|
| Simon128/128 | 128 | 128 | 68 | Feistel | No |
| Speck128/256 | 128 | 256 | 34 | ARX | No |
| ChaCha20 | 512 | 256 | 20 | ARX | No |
| MYSX-256/512 | 256 | 512 | 128 | SPN-like with Feistel cells | No |

MYSX's closest sibling is Simon. The differences (block size, branch organization, transposition, word rotations, non-linear key schedule) are substantial enough that Simon cryptanalysis does not transfer directly (see Section 7).

**Performance.** MYSX is estimated to run at a few MB/s in JavaScript, versus hundreds of MB/s for AES-GCM via WebCrypto (AES-NI). MYSX is not competitive on performance; it is a research primitive.

## 2. Specification

### 2.1. Parameters

| Parameter | Value |
|---|---|
| Block size | 256 bits (8 words of 32 bits) |
| Key size | 512 bits (16 words of 32 bits) |
| Rounds | 128 |
| Structure | SPN-like: 4 parallel Feistel cells + byte transposition |
| Non-linearity | `F(x) = (rotl(x,1) ∧ rotl(x,8)) ⊕ rotl(x,2)` |
| Diffusion | Myszkowski transposition (keyword `MYSZKOWS`) |
| Rotation | `rotl(word, (r mod 31) + 1)` per word and round |
| Final whitening | XOR with 8 words from the key schedule |

**Structural note.** MYSX is sometimes described as a "2-branch Feistel" by analogy with Simon. Strictly, each round applies four independent Feistel cells over 32-bit word pairs, then a byte transposition over the full 32-byte state, then word rotations. It is more accurately classified as an SPN with Feistel cells. The "Feistel" label is a convenience.

### 2.2. Notation

- `rotl(x, n)`: circular left rotation of `x` by `n` bits.
- `∧`: bitwise AND.
- `⊕`: bitwise XOR.
- `s[r]`: state at round `r` (`s[0]` = plaintext, `s[128]` = ciphertext).
- `s[r][i]`: word `i` (0..7) of the state at round `r`.
- `rk[r][i]`: subkey `i` (0..7) of round `r`.
- `rkFinal[i]`: word `i` (0..7) of the final whitening.

## 2.3. Constants

```
CONST[0..7] = [
  0x243f6a88, 0x85a308d3, 0x13198a2e, 0x03707344,
  0xa4093822, 0x299f31d0, 0x082efa98, 0xec4e6c89
]
```

These are the fractional digits of π (same as Blowfish).

### 2.3.1. Myszkowski permutation

Keyword: `MYSZKOWS` (8 letters, S repeated at positions 3 and 8).

Alphabetical order: K(5), M(1), O(6), S(3,8), W(7), Y(2), Z(4).

The byte permutation `MYZ_PERM[0..31]` is constructed as:

```
MYZ_PERM = [
   4,12,20,28,                 # K (column 4)
   0, 8,16,24,                 # M (column 0)
   5,13,21,29,                 # O (column 5)
   2, 7,10,15,18,23,26,31,     # S,S (columns 2 and 7, read row-wise)
   6,14,22,30,                 # W (column 6)
   1, 9,17,25,                 # Y (column 1)
   3,11,19,27                  # Z (column 3)
]
```

The inverse permutation `MYZ_INV` satisfies `MYZ_INV[MYZ_PERM[i]] = i`.

**Design note.** The Myszkowski choice is didactic (classical transposition with a keyword) and adds no cryptographic argument. A byte permutation optimized for diffusion (e.g., a structured permutation over the 4×8 byte grid) would strictly dominate it. We keep Myszkowski in v0.1 because it is the design under review; a future version may replace it.

## 2.4. Key schedule

### 2.4.1. Initialization

Input: key `K[0..63]` (64 bytes).

Initial state `s[0..15]` (16 words of 32 bits):

```
s[i] = big_endian_word(K[4i .. 4i+3])
```

### 2.4.2. Expansion

For `r = 0 .. 127`:

```
spread = (r + 1) * 0x9E3779B9  (mod 2^32)

# Round subkeys
for i in 0..7:
```

```
    rk[r][i] = s[i] ⊕ s[(i+5) mod 16] ⊕ s[(i+11) mod 16]
               ⊕ CONST[(r+i) mod 8] ⊕ spread

# For i = 0..3: fold fk into rk
for i in 0..3:
    fk = s[(i+8) mod 16] ⊕ CONST[(r+i+3) mod 8]
    rk[r][i] = rk[r][i] ⊕ fk

# State update (Feistel 8+8, bijective)
L = s[0..7]
R = s[8..15]
for i in 0..7:
    a = R[i]
    b = R[(i+1) mod 8]
    c = R[(i+3) mod 8]
    F = (rotl((a + b) mod 2^32, 7) ⊕ rotl(c, 13) ⊕ CONST[i] ⊕ spread)
    s[i]     = R[i]
    s[i + 8] = L[i] ⊕ F
```

#### 2.4.3. Final whitening

After the 128 rounds:

```
for i in 0..7:
    rkFinal[i] = s[i] ⊕ s[(i+8) mod 16] ⊕ CONST[(128+i) mod 8]
```

### 2.5. Round function

Each round `r` transforms state `s[r]` into `s[r+1]` in four steps:

**Step 1. AddRoundKey.**

```
for i in 0..7:
    s[r][i] = s[r][i] ⊕ rk[r][i]
```

**Step 2. Feistel cells.** For `p = 0..3`:

```
L = s[r][p]
R = s[r][p + 4]
s_mid[p]     = R
s_mid[p + 4] = L ⊕ F(R)
```

where `F(x) = (rotl(x,1) ∧ rotl(x,8)) ⊕ rotl(x,2)`.

**Step 3. Myszkowski transposition.** Convert `s_mid[0..7]` to bytes (big-endian per word):

```
bytes[4i .. 4i+3] = big_endian(s_mid[i])
permuted[j] = bytes[MYZ_PERM[j]]  for j = 0..31
s_perm[i] = big_endian_word(permuted[4i .. 4i+3])
```

**Step 4. Word rotation.**

```
shift = (r mod 31) + 1
for i in 0..7:
    s[r+1][i] = rotl(s_perm[i], shift)
```

### 2.6. Encryption

```
s[0] = plaintext  (as 8 big-endian words)
for r = 0..127:
    apply round r (Section 2.5)
for i = 0..7:
    s[128][i] = s[128][i] ⊕ rkFinal[i]
ciphertext = s[128]  (as 32 big-endian bytes)
```

### 2.7. Decryption

Inverse of encryption:

```
s[128] = ciphertext
for i = 0..7:
    s[128][i] = s[128][i] ⊕ rkFinal[i]
for r = 127..0:
    apply inverse round r (inverse of Section 2.5, reverse order)
plaintext = s[0]
```

The inverse round consists of:

1. Inverse rotation: `rotl(word, 32 - shift)`.
2. Inverse transposition: `bytes[MYZ_PERM[j]] = permuted[j]`.
3. Inverse Feistel: `R = newL`, `L = newR ⊕ F(R)`.
4. XOR with `rk[r][i]`.

### 2.8. Mode of operation (CTR)

32-byte counter:

```
counter[0..15]  = nonce (16 bytes)
counter[16..31] = 0  (128 bits, big-endian)
```

For each 32-byte plaintext block:

```
keystream = MYSX_encrypt(counter)
ciphertext_block = plaintext_block ⊕ keystream
counter[16..31] += 1  (big-endian)
```

### 2.9. AEAD (encrypt-then-MAC)

Packet: `VERSION(1) ‖ NONCE(16) ‖ CT(n) ‖ TAG(32)`

MAC input: `VERSION ‖ NONCE ‖ AAD_LEN(4 BE) ‖ AAD ‖ CT_LEN(4 BE) ‖ CT`

TAG: `HMAC-SHA256(macKey, MAC input)`

Verification: the decryptor verifies the TAG before decrypting.

### 2.10. Key derivation

**From a random secret (HKDF-SHA256).**

```
encKey = HKDF-SHA256(secret, salt, info = context ‖ ":enc", L = 64)
macKey = HKDF-SHA256(secret, salt, info = context ‖ ":mac", L = 32)
```

**From a password (PBKDF2-SHA256 + HKDF).**

```
prk    = PBKDF2-SHA256(password, salt, iterations = 600000, L = 32)
encKey = HKDF-SHA256(prk, salt, info = context ‖ ":enc", L = 64)
macKey = HKDF-SHA256(prk, salt, info = context ‖ ":mac", L = 32)
```

### 2.11. Test vectors

See `KAT.json` (public). Two digests are documented in `README.md`:

- `KAT.json` public digest: SHA-256 over the full JSON (block + AEAD + KDF). This is the artifact a third-party implementation should reproduce.
- `KNOWN_KAT_DIGEST` (internal): SHA-256 over 16 blocks with `key[i] = i`. Used as fast regression in `mysx.test.js`. It is strictly a subset of the public KAT, not a different KAT.

A third-party implementation should target the **public digest**.

## 3. Differential cryptanalysis

### 3.1. Model

Tool: `misc/mysx_milp.py` (PuLP + CBC/HiGHS). AND model: `a = Δx ∨ Δy`, which counts a cost-1 event whenever at least one input is active. For an AND with independent inputs, this is the exact per-bit cost. The real approximations in the model are:

- The Markov-independence assumption does not hold: the two rotated inputs of F come from the same word `x`.
- We ran a single input difference at r=12–14 (w2b30). Multi-bit input differences may yield lower weights.
- `gapRel=0.2` and `timeLimit=300` return feasible solutions (upper bounds), not certified optima.
- The model captures single trails, not differentials (clusters of trails with the same output difference).

### 3.2. Observed upper bounds

| Rounds | Input | Weight (upper bound) | Single-trail prob. |
|---|---|---|---|
| 5 | w1b24 | 2 | $2^{-2}$ |

| Rounds | Input | Weight (upper bound) | Single-trail prob. |
|---|---|---|---|
| 6 | w1b24 | 6 | $2^{-6}$ |
| 7 | w2b30 | 10 | $2^{-10}$ |
| 8 | w2b30 | 18 | $2^{-18}$ |
| 12 | w2b30 | 68 | $2^{-68}$ |
| 14 | w2b30 | 114 | $2^{-114}$ |

Extrapolation: the 128-bit threshold is reached at r≈15. **This is an extrapolation, not a proof.** It tells us what the model produces for the input we tried; it does not establish the best trail over all inputs, nor that clustering does not lower the effective security.

**Direct verification at r=4 (w=32).** The trail from the table above (w0b0, MILP weight 8, Δout = {w0b4, w1b10, w1b16, w1b30, w6b4}) was verified with 8 fixed keys × $2^{17}$ pairs each, using the full 128-round cipher with the final whitening disabled (to isolate the round function). Observed hit rate: $7.802\times10^{-3}$ over 1,048,576 pairs. Expected from MILP: $2^{-8} = 3.906\times10^{-3}$. **Ratio: 2.00×.**

The trail exists at r=4. The single-trail probability is a factor 2 higher than the MILP predicts, consistent with the clustering factor documented in §3.3. **The MILP model is correct within a factor of 2 at this round count.**

## 3.3. Empirical check (r=4, w0b0)

As described above, the direct verification gives a ratio of 2.00× between the observed probability and the MILP single-trail prediction. Two interpretations are consistent with the data:

- **Differential clustering** (positive): multiple trails with the same output difference Δout contribute to the observed probability, so the effective probability is higher than the single-trail probability.
- **Model systematic overestimate** (less likely): the AND model `a = Δx ∨ Δy` underestimates the per-round cost by a factor that happens to be 2 at r=4. This would imply the true weight is 9, not 8.

Without a full clustering analysis (summing over all trails with the same Δin, Δout), we cannot distinguish these two. The practical implication is the same: the effective weight of the r=4 trail is 7, not 8. **If this factor holds at higher round counts, the "128-bit threshold at r≈15" becomes "at r≈17–18". The margin remains very large, but the extrapolation should be read as approximate.**

## 3.4. What this section does not establish

- A lower bound on the best single-trail weight (requires gap=0 and multi-bit inputs).
- An exact clustering analysis (requires modeling all trails with the same Δout).
- Resistance to related-key differentials.
- That the factor-2 clustering is constant across all round counts.

# 4. Linear cryptanalysis

## 4.1. Overview

An earlier version of this paper (§4 of the previous draft) reported only negative results: three approximate linear MILP models (`misc/mysx_linear_milp.py`) all produced trails that failed empirical verification at w=32. That negative result has now been explained. The models used per-cell AND activity (`act = α ∨ β ∨ γ`) as a proxy for correlation, which is not the correct structure of F's LAT. A correct model exists in closed form and is implemented in `cryptanalysis/models/linear/mysx_linear_model.py` (with detailed documentation in `cryptanalysis/models/linear/LINEAR_MODEL.md` and raw data in `cryptanalysis/models/linear/linear_model_data.json`). The model is machine-checked against the real cipher by `python3 mysx_linear_model.py selftest`.

**Result in one line.** The LAT of F is exactly `corr(b,g) = ±2^{−h(b)}`, where `h(b)` is half the GF(2)-rank of a 32-cycle subgraph determined by `b`; there is a free choice of `g` among `4^{h(b)}` values with sign determined by a quadratic form. Mask propagation is bit-exact; every trail carries an exact key- dependent sign. Verified consequences: `|corr|=1` chains over up to 9 consecutive rounds (three keys), and exact linear-hull predictions at r=10,11,12 that match the real cipher within sampling error ($|z| < 0.4$).

## 4.2. Exact LAT of F

For output mask `b` and input mask `g`,

```
b.F(x) ⊕ g.x  =  Q_b(x) ⊕ l.x ,   l = rotr(b,2) ⊕ g ,
Q_b(x) = ⊕_j b_{j+1} x_j x_{j−7} .
```

`Q_b` is a quadratic form whose graph `G_b` is a subgraph of the 32-cycle `j — j−7` (edge `(j, j−7)` present iff `b_{j+1}=1`; a single cycle because `gcd(7,32)=1`). Let `A_b` be the GF(2) adjacency matrix of `G_b`. Then:

- `corr(b,g) ≠ 0` iff `l = A_b y` for some `y`. There are exactly `2^{rank(A_b)} = 4^{h(b)}` valid `g`.
- For each valid `g`, `corr(b,g) = (−1)^{Q_b(y)} · c_0(b)` with `|c_0(b)| = 2^{−h(b)}`, where `h(b) = rank(A_b)/2`.
- `h(b)` equals the maximum matching = minimum vertex cover of `G_b` (König). The only graph that is not a forest is `b = 0xFFFFFFFF` (full cycle), where `rank = 30` and `h = 15`.
- Parseval holds: `Σ_g corr(b,g)² = 1`.

This is **exact** and **machine-checked** against brute-force LAT of F for random `b` of Hamming weight up to 8 (`selftest`).

**Cost data.**

- Single-bit `b`: `h = 1`, four valid `g`, `|corr| = 1/2`.
- Two-bit `b` (496 masks): histogram `{2: 464, 1: 32}` (the 32 pairs at bit distance 7 are adjacent edges).

- Uniform random 32-bit `b` (16384 samples): histogram of `h` peaks at `h = 11` (4130 samples); full histogram in `linear_model_data.json`.
- Densest masks per unit of weight (bits `(1−7t) mod 32`, `t < 2m`): every extra 2 consecutive edges costs 1; cost per active mask bit approaches 1/2 in the best case, never below 15/32.

### 4.3. Exact mask propagation

For a round `r`, with input mask `m` on `s[r]` and cells `p = 0..3` (`L = s[r][p]`, `R = s[r][p+4]`):

```
b_p     = m[p]                              (only words 0..3 decide the cost)
g_p    ∈ valid_gammas(b_p)          (4^{h(b_p)} choices, each |corr| = 2^{−h(b_p)})
n[p+4]  = m[p] ;  n[p]  = m[p+4] ⊕ g_p
m_next  = lin_fwd(n, r)            (byte transposition, then rotl by (r mod 31)+1)
w_r     = Σ_p h(m[p])
sign    = (Π_p sign_p) · (−1)^{m · rk[r]}
```

Correlation of a trail: `Π_r 2^{−w_r} · (−1)^{Σ_r m_r·rk[r]} · Π_p sign_p`. The linear hull at fixed key is the exact sum over all trails with the same `(m_0, m_R)`; `propagate()` computes it by correlation-vector propagation (pruned by weight). ELP mode (`propagate(..., elp=True)`) gives the key-independent potential under the (incorrect for a deterministic schedule) independent-round-key assumption.

**Free-chain structure.** A round has weight 0 iff words 0..3 of its input mask are zero; the round then acts as a pure bit permutation and the tracked bit never meets `F`. Since the rotation schedule has period 31, there are only 31 distinct windows. The exact free-chain table (all 128 single-bit masks in words 4..7, 31 windows) is in `LINEAR_MODEL.md` §4 and `linear_model_data.json`. Headline: the longest weight-0 chain is **9 consecutive rounds**, at windows `r ≡ 12 (mod 31)` and `r ≡ 27 (mod 31)`.

### 4.4. Verified `|corr| = 1` chains

Each of the 9-round chains has been checked against the real cipher (fixed random keys, $2^{21}$ plaintexts, reduced rounds, final whitening disabled to isolate the round function). Representative results:

| window r0 | rounds | start bit | end bit | key seed | predicted sign | empirical corr | σ |
|---|---|---|---|---|---|---|---|
| 12 | 9 | s[12].174 | s[21].79 | 1 | +1 | +1.00000 | 6.9e-04 |
| 12 | 9 | s[12].174 | s[21].79 | 2 | −1 | −1.00000 | 6.9e-04 |
| 12 | 9 | s[12].174 | s[21].79 | 3 | +1 | +1.00000 | 6.9e-04 |
| 27 | 9 | s[27].146 | s[36].55 | 1 | +1 | +1.00000 | 6.9e-04 |
| 27 | 9 | s[27].146 | s[36].55 | 2 | +1 | +1.00000 | 6.9e-04 |
| 27 | 9 | s[27].146 | s[36].55 | 3 | +1 | +1.00000 | 6.9e-04 |
| 0 | 5 | s[0].208 | s[5].55 | 1 | −1 | −1.00000 | 6.9e-04 |
| 0 | 5 | s[0].208 | s[5].55 | 2 | −1 | −1.00000 | 6.9e-04 |
| 0 | 5 | s[0].208 | s[5].55 | 3 | −1 | −1.00000 | 6.9e-04 |

The magnitude is 1 for every key. Only the sign depends on the key, through `Σ_r m_r · rk[r]`. The model predicts that sign exactly. Per- round masks and complete details are in `linear_model_data.json` (`verified_chains`).

## 4.5. Best single trails beyond the free chain (upper bounds)

Layered beam search with the exact rules, starts = all 256 single-bit masks, beam 1200, cells with `h ≤ 2` expanded. Each reported weight is an **achievable single-trail weight**, i.e. an upper bound on the best weight, not a minimum.

| rounds R | 1–9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|
| best weight, window r0=12 | 0 | 1 | 2 | 3 | 7 | 12 |
| best weight, window r0=27 | 0 | 1 | 3 | 3 | 6 | 11 |

Note that this is *the opposite* of a linear-distinguisher risk at low rounds: at the 9-round mark we still have weight 0 (correlation ±1 in absolute value). The reader should not confuse this with an attack on the full cipher — see §4.7.

## 4.6. Hull verification

The single-trail numbers above are inputs to a hull computation. For three trails at r0 = 12 with weights 1, 2, 3, the model produces an exact signed prediction (fixed key, sum over all trails of weight ≤ w+4 with the same end masks) and the real cipher produces an empirical correlation over $2^{22}$ plaintexts.

| window r0 | R | best-trail w | hull masks kept | predicted corr | empirical corr | σ | z |
|---|---|---|---|---|---|---|---|
| 12 | 10 | 1 | 4 | +0.500000 | +0.499997 | 4.9e-04 | −0.01 |
| 12 | 11 | 2 | 40 | −0.250000 | −0.250008 | 4.9e-04 | −0.02 |
| 12 | 12 | 3 | 3152 | −0.125000 | −0.124813 | 4.9e-04 | +0.38 |

Three independent predictions, three matches, |z| < 0.4 in every case. The hull sum is not an approximation here; it is the exact polynomial expansion of the trail system in the model, and it is what the real cipher produces within sampling error.

## 4.7. What this establishes and what it does not

**Established (machine-checked by `selftest`):**

- Exact LAT of F (valid set, magnitude, sign) against brute force.
- Exact per-round mask rule and key-sign tracking.
- Existence of key-independent-magnitude `|corr| = 1` single-bit relations over up to 9 consecutive rounds.
- Exact signed linear-hull predictions (not approximations) at r=10, 11, 12 against the real cipher.

- The `mysx.prod.js` Python port reproduces `KNOWN_KAT_DIGEST`, so all claims are made against the actual reference implementation.

**Not established:**

- A **lower bound** on the best trail weight over 128 rounds. Beam search gives upper bounds from a specific start distribution.
- An exact 128-round hull. Even the pruned `propagate()` will not scale; a certified bound would need a SAT/MILP encoding.
- The cost of a key-recovery attack.
- Behaviour of multi-bit starts at larger R (only single-bit starts were explored in the beam search).
- Anything about the last-round / whitening structure, related-key linear behaviour, or side channels.

**The weight-0 chains are not an attack.** A 9-round `|corr| = 1` relation sits in a 128-round cipher. Standard linear cryptanalysis extrapolates a weight over the full length; here the trail weight over rounds 0..8 is 0 and the weight over rounds 9..127 is unknown. The result is best read as: *the exact machinery is in place and it agrees with the real cipher on small round counts*, not as *MYSX is broken*. The 128-round picture remains open.

## 4.8. Previous incorrect models and why they failed

For the record, the previous §4 of this paper reported three approximate models, all with negative empirical verification. They used per-cell AND activity as a proxy for correlation, which is not the correct structure of F. The correct structure is the rank of the graph `A_b` above. The previous models produced trails with predicted `|corr| ≈ 2^{−8}` at `r=4` but no detectable signal — consistent with the fact that those trails have zero true correlation in the exact model.

An instructive detail from the old work remains relevant to *any* future linear work on MYSX: the empirical verification harness must use **fixed keys** per sample. With random keys per plaintext, the sign of every linear trail is key-dependent and the observed correlation of a genuinely nonzero trail collapses to zero. See `misc/validate_lineal_harness.js`, which validates the harness against a hand-computed `w=8` case (observed correlation 0.4993 vs 0.5).

## 4.9. Integral / division property (preliminary, limited)

Tool: `misc/mysx_division_property_v2.py` with standard division- property rules (COPY: `k_a = Σ k_copies`; XOR: `k_c = k_a + k_b` with `(1,1)` invalid; AND: `k_c = k_a OR k_b`). The XOR and OR variants were found to be equivalent for all input patterns tested (the `(1,1)` constraint never fired), so the reported results do not depend on the choice between them.

**Results.**

| Input | Active bits | Integral size | Rounds with balanced bits | Saturation |
|---|---|---|---|---|
| bit 0 of word 0 | 1 | 2 | 10 | R = 11 |
| word 0 | 32 | $2^{32}$ | 6 | R = 7 |
| words 0 and 4 | 64 | $2^{64}$ | 5 | R = 6 |

The **integral distinguishers** are the last two rows: 6 rounds with a $2^{32}$-text integral, 5 rounds with a $2^{64}$-text integral. These cover 4.7% and 3.9% of the 128 rounds, leaving 122 and 123 rounds without an integral distinguisher in the tested pattern.

The first row (1 active bit, 2-text integral) is **not** a meaningful integral distinguisher: with 2 texts, "balanced output bit" is equivalent to "output bit independent of the input bit". It is reported as a **single-bit dependency propagation** result — it measures how many rounds are needed for the active bit to reach all 256 output bits (10) — but no integral security conclusion is drawn from it.

**Empirical verification.** The 6-round distinguisher (word 0 active) and the 10-round single-bit dependency were verified against the real cipher (fixed random key, all non-active input bits random) over 20,000 samples each. The balanced bits predicted by the model never flipped; control bits (predicted not balanced) flipped with probabilities 6.5%–87%. The 5-round distinguisher (words 0,4 active) was not verified empirically (its integral has $2^{64}$ texts; a random- sample approximation with 20,000 samples is not meaningful).

**Comparison with Simon (with care).** Simon32/64 has an 11-round integral distinguisher with 31 active bits (34% of its 32 rounds; Todo–Morii, FSE 2016). In **relative terms** MYSX resists integral distinguishers better (4.7% vs 34% coverage), which is consistent with MYSX's more aggressive diffusion (4 parallel Feistel cells + transposition). In **absolute terms** Simon's distinguisher reaches further (11 vs 6 rounds) because MYSX has 4× more rounds. The two are not directly comparable without normalizing by total rounds.

**What this establishes and what it does not.**

Established:

- MYSX has integral distinguishers covering 5–6 rounds with multi-bit inputs ($2^{32}$ and $2^{64}$ texts).
- A single active input bit stops influencing all 256 output bits after 10 rounds.
- The standard division-property rules with the `(1,1)`-invalid XOR constraint produce the same results as the OR-conservative rule for the tested inputs.

Not established:

- Integral distinguishers beyond 6 rounds for multi-bit inputs.
- Behaviour for r0 ≠ 0. Only r0 = 0 (the first round window) was tested. The rotation schedule has period 31, and windows r0 = 12 and r0 = 27 are where the linear chains are longest; they may also give longer integral distinguishers, or shorter. Not explored.
- Full-block inputs (255 active bits).

- Three-subset DP (Hu et al. 2021) or traceability-enhanced DP (Wang et al. 2023), which may extend the distinguishers.
- Any key-recovery attack based on these distinguishers.

**Historical note.** The XOR rule of Todo–Morii as commonly written in the literature (`k_c = k_a XOR k_b`) is not sound when the two XOR operands share variables. In MYSX's `F = C ⊕ D`, both `C` and `D` depend on `R`, so that rule overestimates balanced bits via spurious cancellations. The first implementation of this analysis used that rule and produced oscillating, non-converging numbers (127 → 117 → 152 → ...). Those numbers are an artifact and are not reported here. The results above use the standard `(1,1)`-invalid rule, which is sound.

## 5. Related-key cryptanalysis

Tools: `misc/keyschedule_test.js` (basic) + `misc/keyschedule_test_extended.js` (22 tests). Coverage: exhaustive weight-1 (512 bits), exhaustive weight-2 (130,816 pairs), weight-3 "same Δ" (1,680 trials), beam search up to weight-6 (beam 50), 5 base keys, structured patterns, cycles, weak keys.

**Results (summary):**

- No persistent `Δrk[r]=0` for r≥1 in any test.
- Beam search score: 139 (ideal 128).
- No weak keys, cycles, or fixed points.
- Slow diffusion at r=1–4 (expected: one Feistel round per cipher iteration).
- Worst bit (minimum weight at r=1): bit 160 (word 5, bit 0), 1-bit difference. Consistent across 5 base keys (structural, not noise).

**Limitations.** These are heuristic tests, not a proof. We did not run a targeted MILP/SAT related-key model (~650k variables); adaptive related keys, biclique attacks, and algebraic attacks on the key schedule are not covered. **MYSX's related-key security is unknown.**

## 6. Generic attacks

| Attack | Complexity | Verdict |
|---|---|---|
| Classical brute force | $2^{512}$ operations | Infeasible |
| Grover (quantum) | $2^{256}$ iterations | Infeasible with foreseeable technology |
| Block collisions (birthday) | $2^{128}$ blocks | Infeasible |

**Note on Grover.** With a 512-bit key, Grover leaves 256 bits of effective security. With 1,000,000 logical qubits at 1 ns/iteration, the time would be ~$10^{60}$ years. This is a theoretical estimate; no post-quantum analysis of the cipher beyond Grover has been done.

**Note on the 512-bit key.** HKDF-SHA256 produces at most a 256-bit pseudorandom key. If the master secret has 256 bits of entropy, the effective security is 256 bits, not 512. The "key size" of MYSX is not the security level of the system.

# 7. Simon cryptanalysis: what transfers and what does not

MYSX uses Simon's `F`. It is **not** Simon. The differences:

- Block size and branch organization: MYSX is an SPN-like structure with four parallel Feistel cells over 32-bit word pairs (Section 2.1). Simon is a classical 2-branch Feistel over `n`-bit halves.
- Four parallel `F` per round (Simon uses 1).
- Myszkowski byte transposition (Simon has none).
- Round-dependent word rotations (Simon has none).
- Non-linear ARX key schedule (Simon's is linear).
- 128 rounds vs 68.

**What transfers.** The exact LAT of `F` — its closed form `|corr| = 2^{-h(b)}`, its validity condition `l ∈ colspace(A_b)`, and its sign structure — is a property of the function and therefore applies to MYSX regardless of the surrounding structure (Section 4.2 is the transfer). That is the correct starting point for MYSX-specific linear cryptanalysis; the failed models of the previous draft used a different (incorrect) F model.

**What does not transfer.** Key recovery from Simon's linear key schedule does not transfer: MYSX's key schedule is non-linear. Linear hull attacks against Simon are highly structured around Simon's Feistel-branch and its round-reduced behaviour; the MYSX hull is a different object and must be computed from the MYSX round model (Section 4.6). Integral distinguishers against Simon rely on Simon's particular 2-branch Feistel structure and its key schedule; the preliminary MYSX integral distinguishers in §4.9 are considerably shorter, and their relationship to Simon's is one of normalized coverage, not direct transfer.

**Relevant literature and published attacks on Simon128/128** (68 rounds):

- Linear-hull distinguishers covering ~49 rounds, complexity $\sim 2^{127}$ (Chen–Wang, "Improved linear hull cryptanalysis of round-reduced versions of SIMON", ePrint 2014/973).
- Differential distinguishers covering ~46–50 rounds, complexity $\sim 2^{125.7}$ (Biryukov–Roy–Velichkov, "Differential analysis of block ciphers SIMON and SPECK", CRYPTO 2015).
- SAT-based algebraic distinguishers covering ~12–14 rounds (practical on a PC).
- Integral distinguishers covering ~26 rounds for Simon128/128 with multi-bit inputs, and ~11 rounds for Simon32/64 with 31 active bits (Todo–Morii, FSE 2016).
- Structural observations on the SIMON family (Kölbl–Leander–Tiessen, "Observations on the SIMON block cipher family", CRYPTO 2015).

None of these results is transferred to MYSX in this paper. The MYSX-specific analysis (Sections 3, 4, 5) is what provides the evidence; Simon attacks are context, not direct transfer.

# 8. Analysis limitations

- **Differential**: upper bounds on single-trail for one input at r=12–14; one trail verified at r=4 (ratio 2.00×); no lower bounds; no full clustering analysis.
- **Linear**: an exact model of `F` is established and matches the real cipher on small round counts, including exact signed hulls at r=10, 11, 12. No lower bound over 128 rounds; only upper bounds from single-bit starts in the beam search; no key-recovery cost estimate; multi-bit starts unexplored.
- **Integral**: preliminary. 6-round distinguisher with $2^{32}$ texts (word 0 active) and 5-round distinguisher with $2^{64}$ texts (words 0,4 active), both with the sound `(1,1)`-invalid division-property rule. A 10-round single-bit dependency propagation is reported for context but is not an integral distinguisher. Only r0 = 0 tested; other round windows, full-block inputs, and three-subset / traceability- enhanced DP are not explored. No key-recovery attack was attempted.
- **Related-key**: heuristic tests up to weight-6; no targeted MILP.
- **Impossible differential, boomerang, algebraic**: not analyzed.
- **No independent public cryptanalysis.**
- **No post-quantum analysis** beyond the trivial Grover bound.
- **No side-channel analysis.**
- **Functional tests** (`mysx.test.js`, 16/16 PASS) cover round-trip, AEAD negatives, nonce policy, PBKDF2, avalanche. They do not prove security.

# 9. Engineering notes

## 9.1. API surface

- Public API: `aeadEncrypt`, `aeadDecrypt`, `deriveAeadKeys`, `deriveAeadKeysFromPassword`, `NonceGuard`.
- Test API: **disabled by default**. Enabled only when the environment variable `MYSX_ENABLE_TEST=1` is set in the process before requiring the module. There is no code-level toggle.
- CTR (`_ctrXor`) is not exported from the public API.

## 9.2. NonceGuard

`NonceGuard` is a best-effort defense against nonce reuse. It:

- counts messages per key (default $2^{32}$);
- detects recent nonce reuse (window of 1024 per key);
- caps the number of tracked keys (default 1024, LRU eviction);
- identifies keys by a non-cryptographic 64-bit hash (FNV-1a variant), **not** by the raw key bytes; `onSave(state)` therefore does not leak the secret key;
- throttles `onSave` calls (default: once every 100 `check()` calls) to avoid per-message state copies;
- **does not persist across restarts by itself**;
- **does not share state across processes or devices**.

Persistence is provided by `onSave(state)` and `initialState` at construction, or by

`save()` / `load(state)` for manual control.

**It is not a substitute for key lifecycle management.** If the application reuses `(key, nonce)` across restarts, CTR confidentiality is broken. See `mysx.prod.js` for the exact contract.

### 9.3. aeadEncrypt size validation

`aeadEncrypt` validates plaintext size **before** allocating the CTR output buffer, and enforces a configurable `MAX_PLAINTEXT_BYTES` limit (aligned with `MAX_CT_BYTES`, $2^{31}-1$). Applications that require larger limits should raise both limits explicitly and manage memory accordingly.

### 9.4. Type validation

All public functions validate that byte inputs are `Uint8Array` of the expected length (where applicable). Passing a `string` as `aad` or `plaintext` throws instead of silently coercing to array-like behavior.

### 9.5. KAT digests

Two digests are documented in `README.md`:

- `KAT.json` public digest: SHA-256 over the full JSON (block + AEAD + KDF). This is the artifact a third-party implementation should reproduce.
- `KNOWN_KAT_DIGEST` (internal): SHA-256 over 16 blocks with `key[i] = i`. Used as fast regression in `mysx.test.js`. It is strictly a subset of the public KAT, not a different KAT.

A third-party implementation should target the **public digest**.

### 9.6. Harness validation

The linear sampling harness is validated against a hand-computed case at w=8 (r=1, α = {word0 bit2, word4 bit0}, β = word0 bit3, fixed key). The linear approximation $x_0 \oplus F(x)_2 = x_1 x_6$ has correlation exactly ±1/2 with fixed key. Observed correlation: **0.4993** (expected 0.5, within sampling noise). With random keys per sample, the sign of the correlation flips per sample and the observed |corr| drops to **0.0022** (expected ≈ 0).

This confirms the harness is correct when used with fixed keys. It also confirms that the practice of "random keys per sample", used in an earlier version of `misc/mini_trail.js`, washes out genuine linear signals and must not be used for linear cryptanalysis.

See `misc/validate_lineal_harness.js`.

The mini-MYSX with `w=8, kw='MYSZKOWS'` is **degenerate**: the permutation becomes `[4,0,5,2,7,6,1,3]`, the L→R→L cycle at word 0 means a single-bit difference never

passes through F, and the differential probability is 1 at all rounds. This is not representative of MYSX. Use `w=8, kw='MYSZ'` (non-degenerate from r=3) or w=32 directly. See `misc/check_mini_nondegenerate.js`.

The integral division-property harness (`misc/verify_division.py`) is validated against the R=10 single-bit dependency: the two bits the model predicts as balanced (word 0 bit 31, word 3 bit 6) are confirmed balanced over 20,000 random samples, and the other 62 bits of words 0 and 3 are confirmed not balanced. See `misc/verify_division.py` and §4.9.

## 10. Verdict

**MYSX-256/512 v0.1-draft is an experimental proposal, not a secure cipher.** The differential analysis reports upper bounds on single-trail weights, not a security proof; one trail at r=4 has been directly verified with clustering factor 2.00×. The linear analysis now has a correct, exact model of `F` and of the round; it reproduces the real cipher's linear behaviour on the small round counts where it has been checked (verified `|corr| = 1` chains at 5 and 9 rounds; exact signed hulls at r = 10, 11, 12). That is an input to cryptanalysis, not a bound. No lower bound over 128 rounds has been established, no key- recovery attack has been attempted, and multi-bit starts at high round counts are unexplored. The integral analysis is preliminary: it establishes 5–6 round integral distinguishers with multi-bit inputs ($2^{32}$ and $2^{64}$ texts) and a 10-round single-bit dependency propagation, but only for the first round window (r0 = 0). The related-key analysis is heuristic. No impossible-differential, boomerang, or algebraic analysis has been done. The cipher has not been externally audited.

**Do not use MYSX to protect real data.** For production: **AES-256-GCM** or **XChaCha20-Poly1305**.

## 11. Reproducibility

| File | Purpose |
|---|---|
| `SPEC.md` | Formal specification (identical content to Section 2) |
| `PAPER.md` | This document |
| `KAT.json` | Test vectors (block, AEAD, KDF) |
| `generate_kat.py` | KAT generator |
| `mysx.prod.js` | Implementation (public API) |
| `mysx.test.js` | Functional tests |
| `cryptanalysis/models/linear/mysx_linear_model.py` | Exact linear model of F + round; `selftest` / `report` commands |
| `cryptanalysis/models/linear/LINEAR_MODEL.md` | Detailed documentation of the exact linear model |
| `cryptanalysis/models/linear/linear_model_data.json` | All data in Section 4 as JSON |
| `misc/mysx_milp.py` | Differential MILP |

| File | Purpose |
|---|---|
| misc/mysx_linear_milp.py | Previous approximate linear MILPs (historical, superseded by mysx_linear_model.py) |
| misc/mysx_division_property.py | Bit-based division-property MILP (v1; --xor-rule selects XOR or OR) |
| misc/mysx_division_property_v2.py | Bit-based division-property MILP (standard (1,1)-invalid XOR rule) |
| misc/verify_division.py | Empirical verification of integral distinguishers |
| misc/keyschedule_test.js | Basic key schedule audit |
| misc/keyschedule_test_extended.js | Exhaustive audit (22 tests) |
| misc/check_trail.js | Empirical differential trail at r=4 |
| misc/check_linear_trail2.js | Empirical check of the superseded linear model (negative result) |
| misc/validate_lineal_harness.js | Hand-computed linear harness check |
| misc/mini.js | Mini-MYSX (w=8, w=16, w=32) |
| misc/check_mini.js | Mini-MYSX validation vs mysx.prod.js |
| misc/check_mini_nondegenerate.js | Keyword/w compatibility check |
| misc/mini_linear_fast.js | Linear sampling at w=8, kw='MYSZ' |
| misc/verify_trail_w32.js | Per-trail verification at w=32 with MILP weight |
| misc/avalanche.js | Diffusion and avalanche ladder (r=1..24) |

Fixed seed in tests (0xC0FFEE). MILPs are deterministic given the solver and gapRel. The mini-MYSX with w=32 reproduces mysx.prod.js exactly (F function and MYZ_PERM are bit-identical, verified in misc/check_mini.js). The exact linear model's selftest additionally verifies the Python port against KNOWN_KAT_DIGEST of mysx.prod.js, so the linear results are stated relative to the reference implementation.

**Division-property results (§4.9).** All three input patterns (single bit, word 0, words 0+4) were run with --r0 0. The (1,1)-invalid XOR rule and the OR-conservative rule were checked to agree on the tested patterns. Per-round balanced-bit counts are in the terminal logs of the runs; the runtimes were 0.4 s to 4.8 ks with SCIP. The R=10 single-bit dependency was verified against the real cipher (20,000 random samples per bit). The R=6 word-0 integral was verified against the real cipher at the last round (R=6, balanced bits confirmed; R=7 saturated). The R=5 words-0,4 integral was not verified empirically because its $2^{64}$-text integral cannot be approximated by 20,000 random samples.

## 12. Final warning

**MYSX has not been externally audited.** This paper is the author's self-analysis, not a certification. No result here replaces an independent cryptographic review.

**Do not use MYSX to protect real data.**